



Lab 01: Securing VMs using Chrome Enterprise Premium

Build a Secure Google Cloud Network | GSP1036

Overview

Used Chrome Enterprise Premium (BeyondCorp) and Identity-Aware Proxy (IAP) TCP forwarding to enable secure administrative access to VM instances that have no external IP addresses and cannot be accessed directly from the internet.

Key Concepts

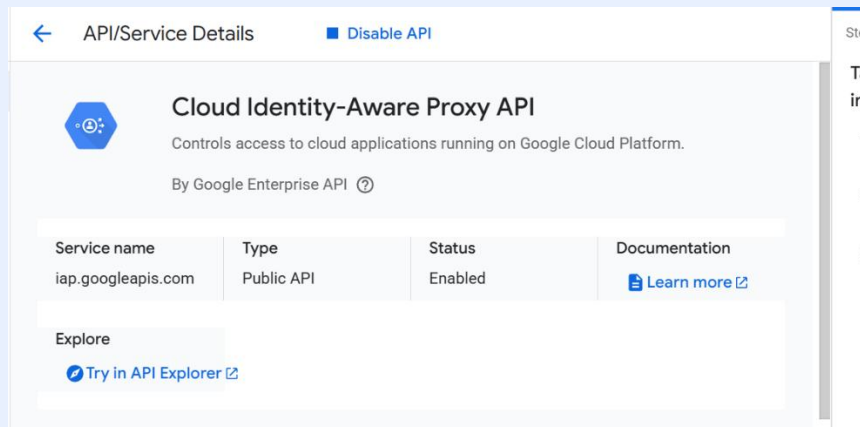
Term	Definition
IAP	Identity-Aware Proxy - controls access to cloud applications and VMs based on identity
IAP Tunnel	Encrypted tunnel that allows SSH/RDP to VMs without external IPs
BeyondCorp/BCE	Zero trust security model that verifies identity before granting access
IAP Desktop	GUI tool that enables RDP and SSH access through IAP tunnels

Objectives

- Enable IAP TCP forwarding in the GCP project
- Create Linux and Windows VM instances with no external IPs
- Configure firewall rules for IAP access
- Grant IAP tunnel permissions via IAM
- Use IAP Desktop to connect to Windows and Linux instances
- Demonstrate SSH and RDP tunneling via IAP

Task 1 - Enable IAP TCP Forwarding

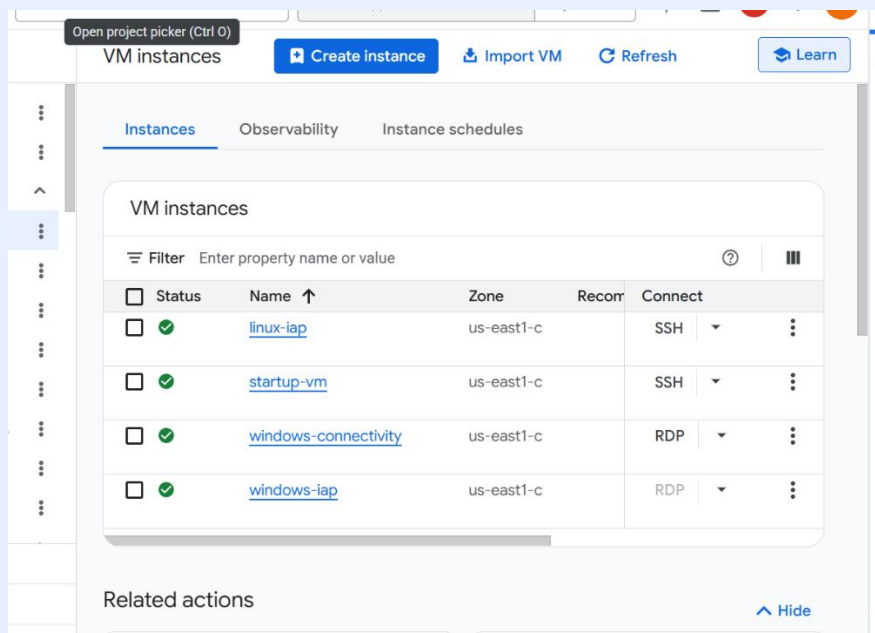
Enabled the Cloud Identity-Aware Proxy API via APIs and Services → Library.



Task 2 - Create Linux and Windows Instances

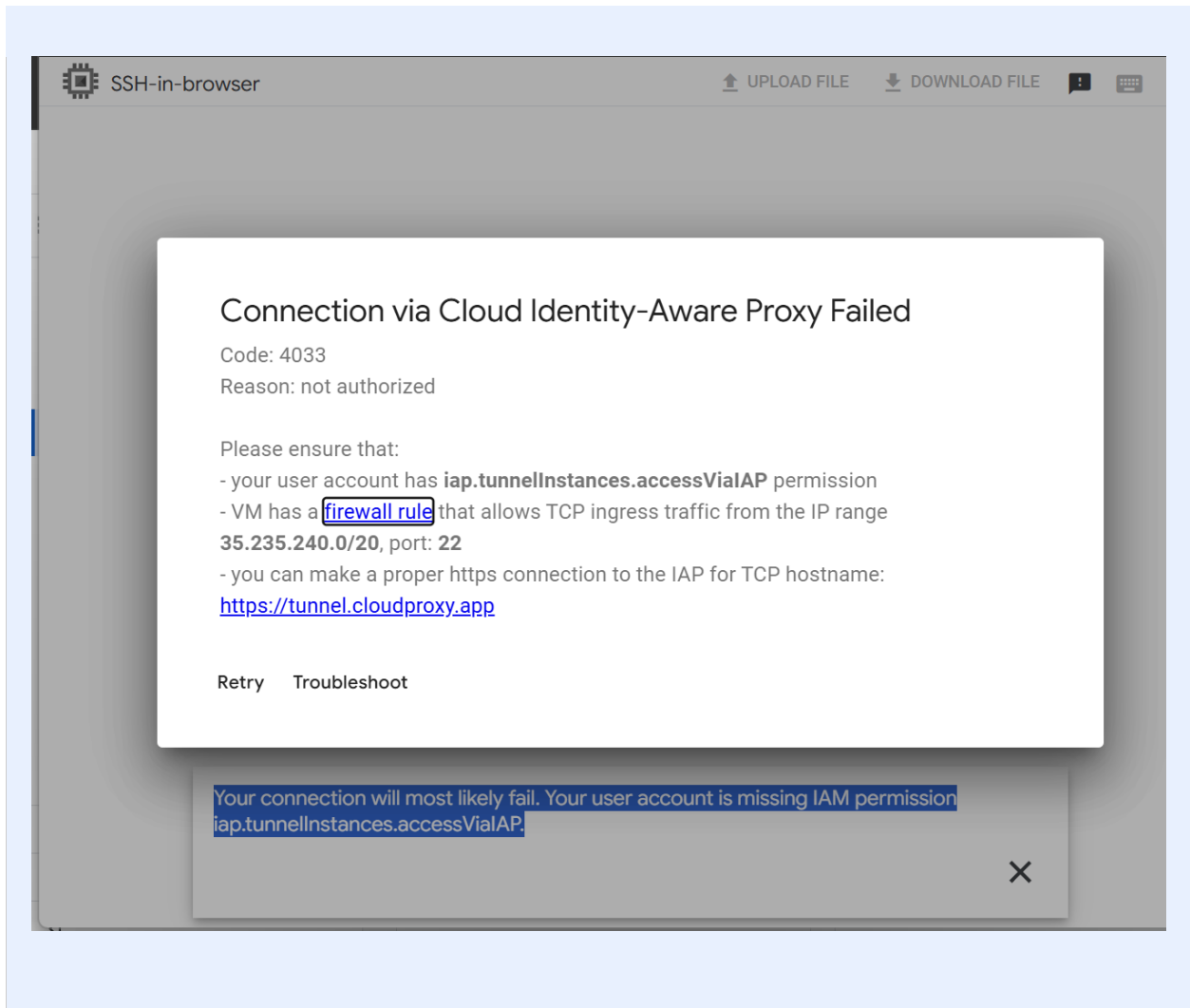
Created 3 VMs: linux-iap and windows-iap with NO external IPs, and windows-connectivity with an external IP for testing.

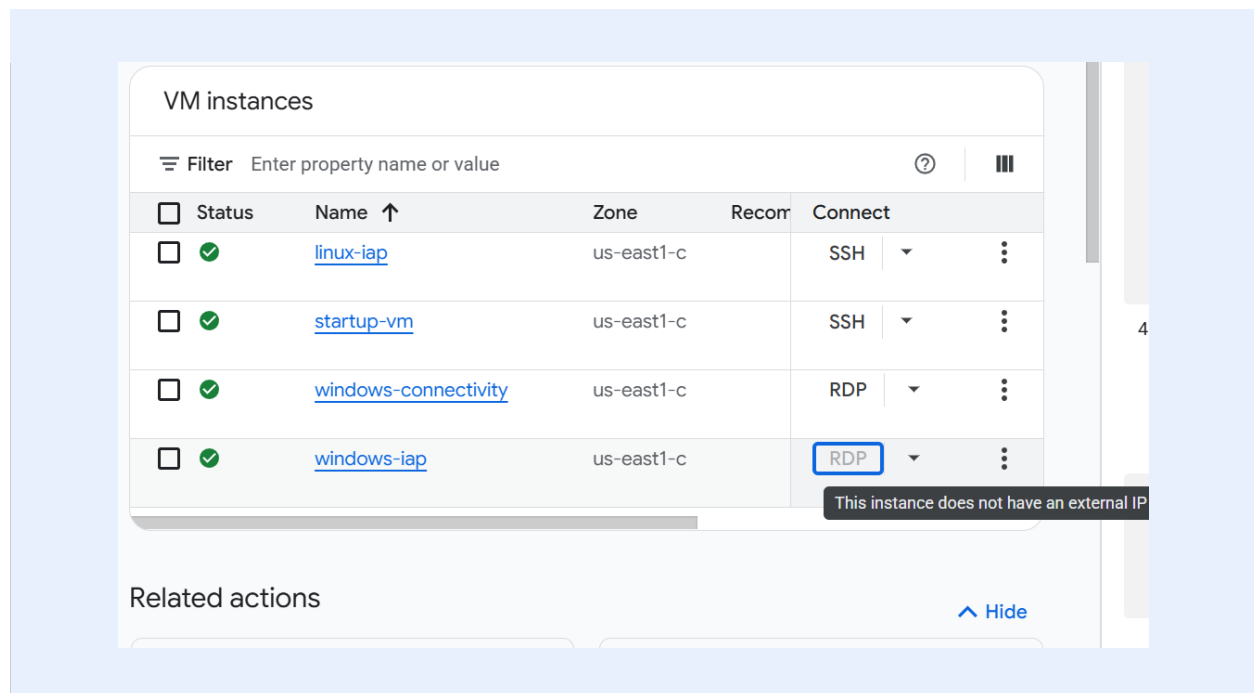
- linux-iap: No external IP, used for SSH tunnel testing
- windows-iap: Windows Server 2016, no external IP, used for RDP tunnel testing
- windows-connectivity: Has external IP, used as the jump box with IAP Desktop installed



Task 3 - Test Connectivity Without IAP

Confirmed that linux-iap and windows-iap cannot be accessed directly because they have no external IP addresses.

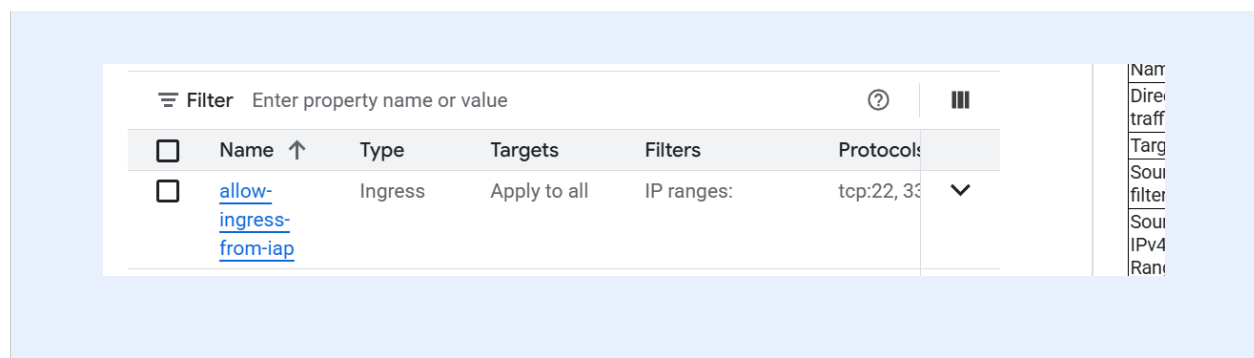




Task 4 - Configure Firewall Rules for IAP

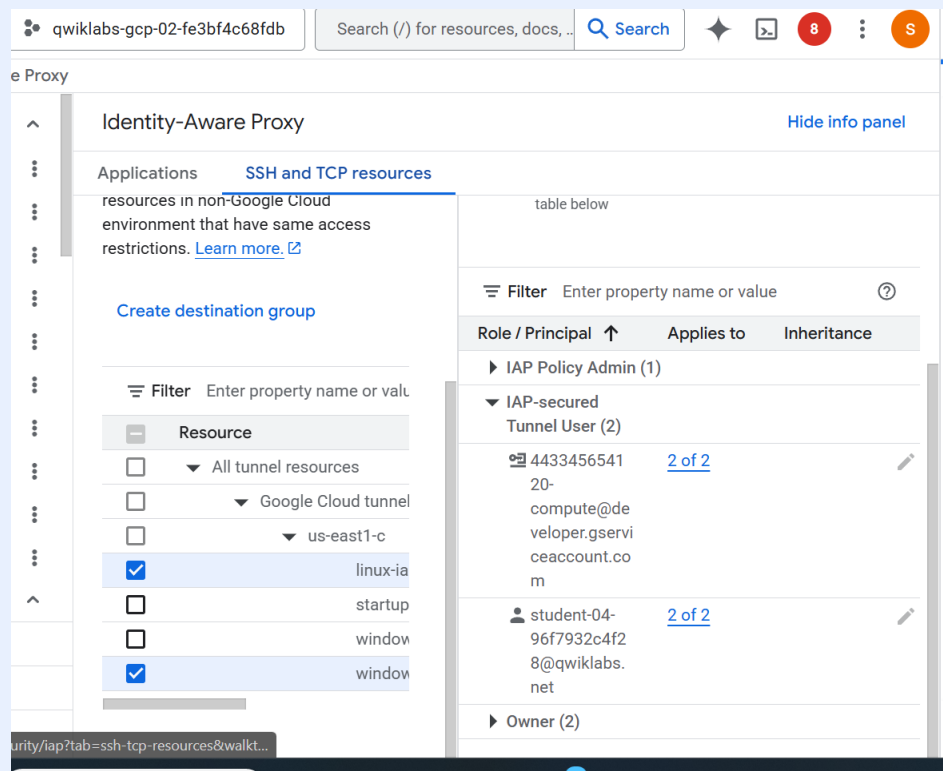
Created a firewall rule to allow IAP traffic from Google's IAP IP range.

- Rule name: allow-ingress-from-iap
- Direction: Ingress
- Source IP ranges: 35.235.240.0/20 (Google IAP range)
- Protocols: TCP ports 22 (SSH) and 3389 (RDP)



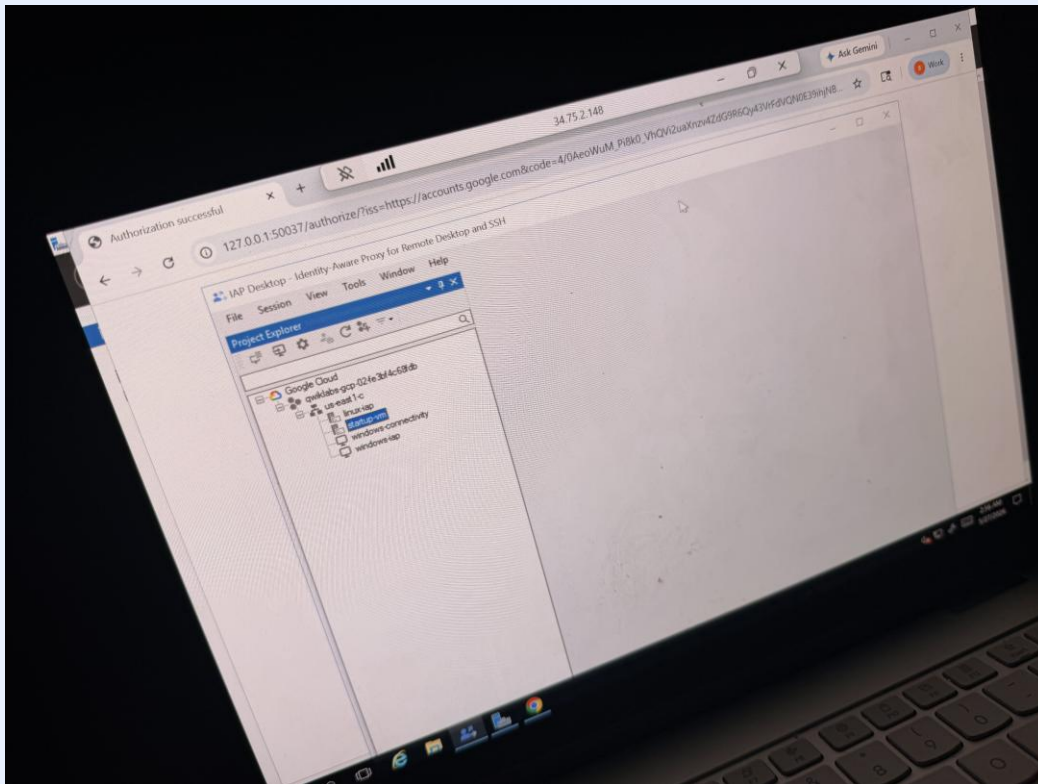
Task 5 - Grant IAP Permissions

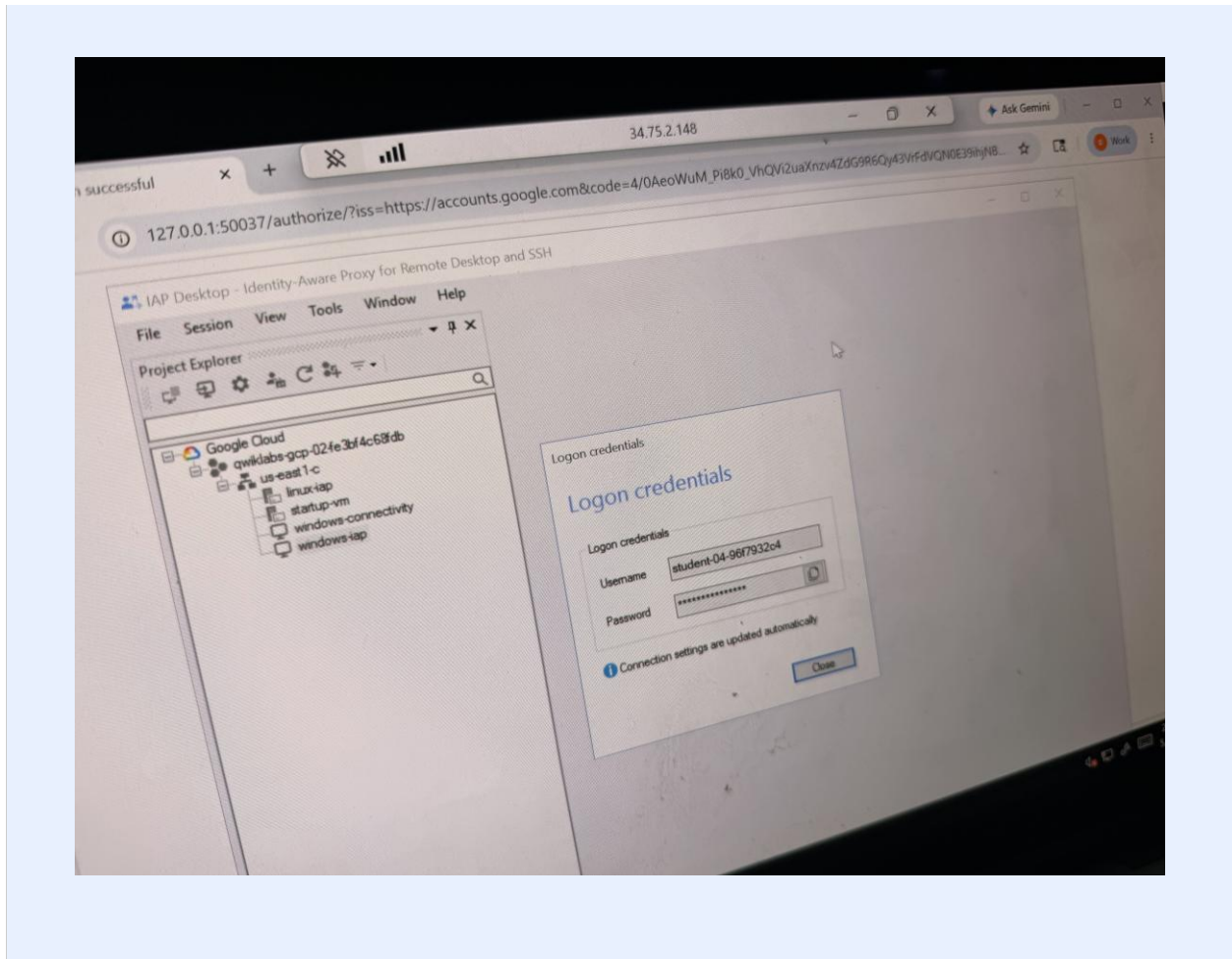
Granted the IAP-Secured Tunnel User role to the windows-connectivity service account and the student account via Security → Identity-Aware Proxy.



Task 6 - Use IAP Desktop

RDP'd into windows-connectivity, opened IAP Desktop, signed in with Google credentials, added the project, and connected to windows-ia through the IAP tunnel.





Task 7 - SSH and RDP Tunneling

From windows-connectivity Cloud SDK Shell, SSH'd into linux-iap and created an IAP tunnel to RDP into windows-iap.

```
gcloud compute ssh linux-iap
gcloud compute start-iap-tunnel windows-iap 3389 --local-host-port=localhost:0 --
zone=[ZONE]
```


Real World Application

- Banks use IAP to give engineers access to internal servers without exposing them to the internet
- Healthcare companies use IAP tunnels to access patient data systems securely
- Zero trust model means access is verified by identity not network location

What I Learned

- VMs without external IPs cannot be accessed directly from the internet
- IAP tunnels allow SSH and RDP access without exposing VMs publicly
- The IAP source IP range 35.235.240.0/20 must be allowed in firewall rules
- IAP Desktop provides a GUI for managing IAP connections on Windows
- BeyondCorp zero trust model verifies identity before granting any access

Resources

GCP IAP Docs: <https://cloud.google.com/iap/docs>

Lab ID: GSP1036